

a strategic principal who wants to persuade agents of an incorrect belief can distort the learning process by leveraging social connections and echo chambers to propagate misinformation. Similarly, models of misinformation “contagion”— without Bayesian agents or strategic decisions — have been studied in [Budak et al. \(2011\)](#), [Nguyen et al. \(2012\)](#), and [Törnberg \(2018\)](#). Our contribution relative to this literature is the possibility that misinformation spreads because of the strategic interactions of Bayesian agents and is exacerbated by profit-maximizing platform algorithms.

There is a growing literature on information design by platforms, building for the most part on the concept of Bayesian persuasion ([Kamenica and Gentzkow \(2011\)](#) and [Kamenica \(2019\)](#)). [Candogan and Drakopoulos \(2020\)](#) study how a platform with private knowledge of content’s accuracy should optimally signal to rational users whether to engage with it, while [Chen and Papanastasiou \(2021\)](#) and [Keppo et al. \(2019\)](#) consider more manipulative actions by platforms, including strategic seeding of information or “cheap talk” signals about quality. Also related are works on reputation and media bias. Motivated by the 2016 presidential election, [Allcott and Gentzkow \(2017\)](#) study the incentives of certain outlets to present misleading news, while [Gentzkow and Shapiro \(2006\)](#), [Hsu et al. \(2020\)](#) and [Allon et al. \(2021\)](#) explore other strategic reasons for media bias. Our paper contributes to this literature by highlighting the role of ideological leaning, strategic interactions, ideological homophily, and platform algorithms.

The most closely related work to ours is [Papanastasiou \(2020\)](#), who studies a model where agents hold heterogenous ideological beliefs and digest (and potentially share) a news article sequentially. Our work is different in three important dimensions. First, [Papanastasiou \(2020\)](#) focuses on costly inspection, which makes sharing decisions strategic substitutes, while our model generates strategic complementarities, because individuals care about further shares of the content they share.⁵ All of our results and formal analysis turn on strategic complementarities. Second, and relatedly, echo chambers play no role in [Papanastasiou \(2020\)](#).⁶ Third, our analysis of engagement-maximization by the platform and its implications for the spread of low-reliability content has no counterpart in [Papanastasiou \(2020\)](#) or any other work in this area we are aware of.⁷

The rest of the paper is organized as follows. The next section introduces our basic environment and describes the information structure and payoffs. Section 3 characterizes the (Bayesian-Nash) equilibria of this model and provides some basic comparative static results. Section 4 studies the effects of homophily by focusing on a special class of sharing networks that correspond to a set of “islands” of like-minded individuals who are less closely linked to those in other islands. Section 5 endogenizes the sharing network as a result of the algorithmic choices of the platform that aims to maximize engagement. Section 6 discusses a range of regulations aimed at containing misinformation.

⁵Our reading of the evidence is that strategic complementarities are more relevant for social media behavior than strategic substitutabilities. For example, [Eckles et al. \(2016\)](#) find evidence that feedback or “encouragement” from peers about Facebook posts have contributed significantly to future behavior and posting. See also [Taylor and Eckles \(2018\)](#) and [Aral and Dhillon \(2018\)](#).

⁶As already noted, echo chambers appear central to the spread of misinformation in practice. See, for example, [Lee et al. \(2011\)](#), [Törnberg \(2018\)](#), [Centola \(2010\)](#), and [Centola and Macy \(2007\)](#)

⁷[Papanastasiou \(2020\)](#) also discusses platform incentives, but assumes that the platform is interested in limiting misinformation. Our reading of the evidence in this instance, too, favors our interpretation, where platforms such as Facebook are (or at the very least used to be before regulatory pressure mounted) fairly indifferent to the presence of misinformation but strongly prioritize engagement maximization.